

ICD-9-CM codes with no match in the target system — 27 August 2026

There are 9 ICD-9-CM codes with no ICD-10-CA match and 52 with no ICDA-8 match. The algorithm uses two numbers to decide: how often a pair of codes appears together in patient records, and the cosine similarity between the two code labels. Similarity here is from ClinicalBERT.

ICD-9-CM to ICD-10-CA, the 9 codes

Co-occurrence, all 9. Partners is the number of different ICD-10-CA codes the code shares a record with, and the columns after it describe the counts across those partners. The source data does not report a count below 7.

Code	Label	Partners	Min	25th	Median	75th	Max	Most frequent partner
338	pain not elsewhere classified	827	7	18	49	181.5	13,431	E11, Type 2 diabetes mellitus
515	postinflammatory pulmonary fibrosis	306	7	13	24	53.8	5,194	J84, Other interstitial pulmonary diseases
239	neoplasms of unspecified nature	585	7	14	34	91	4,862	I10, Essential (primary) hypertension
327	organic sleep disorders	349	7	12	23	49	3,108	G47, Other sleep disorders
249	secondary diabetes mellitus	161	7	12	18	38	1,783	E11, Type 2 diabetes mellitus
209	neuroendocrine tumors	124	7	11.8	17	31.8	506	C78, Secondary malignant neoplasm of respiratory and digestive organs
175	malignant neoplasm of male breast	24	7	8	10.5	22.5	131	C79, Secondary malignant neoplasm of other and unspecified sites
339	other headache syndromes	58	7	9.2	15	17.8	54	E11, Type 2 diabetes mellitus
445	atheroembolism	2	7	9.2	11.5	13.8	16	I74, Arterial embolism and thrombosis

These counts are not low. Code 338 shares a record with 827 different ICD-10-CA codes, up to 13,431 times. Across the 345 codes that do have a match the middle value of the highest count is 1,288.5, so this group sits above it rather than below. Only 445 and 339 are low.

Cosine similarity, each code scored against all 2038 ICD-10-CA codes.

Code	Label	Average	Min	25th	Median	75th	Max	Closest target code
175	malignant neoplasm of male breast	0.825	0.641	0.796	0.826	0.849	0.955	C30, Malignant neoplasm of nasal cavity and middle ear
249	secondary diabetes mellitus	0.817	0.663	0.774	0.817	0.859	0.951	O24, Diabetes mellitus in pregnancy
239	neoplasms of unspecified nature	0.831	0.662	0.804	0.833	0.859	0.942	D15, Benign neoplasm of other and unspecified intrathoracic organs
515	postinflammatory pulmonary fibrosis	0.852	0.694	0.823	0.858	0.886	0.932	N72, Inflammatory disease of cervix uteri
209	neuroendocrine tumors	0.836	0.693	0.808	0.842	0.867	0.915	A74, Other diseases caused by chlamydiae
327	organic sleep disorders	0.770	0.541	0.725	0.773	0.823	0.915	P78, Other perinatal digestive system disorders
339	other headache syndromes	0.816	0.648	0.786	0.820	0.851	0.915	A69, Other spirochaetal infections
338	pain not elsewhere classified	0.805	0.685	0.785	0.807	0.827	0.893	R52, Pain, not elsewhere classified
445	atheroembolism	0.711	0.552	0.686	0.713	0.736	0.853	M66, Spontaneous rupture of synovium and tendon

The highest score for these 9 runs from 0.853 to 0.955, against 0.861 to 0.981 for the 345 codes that do have a match. The closest ICD-10-CA code is clinically unrelated in 8 of the 9: code 175, malignant neoplasm of male breast, is closest to C30, malignant neoplasm of nasal cavity and middle ear. The exception is code 338, whose closest target R52 carries the same label, yet the crosswalk records no match.

ICD-9-CM to ICDA-8, the 52 codes

None of the 52 appear in the co-occurrence data at all, so there is nothing to describe for that signal. That is why the algorithm returns no match for them, and also why it returns no match for 39 codes that do have one, since 32 of the 302 matched codes are missing from the same data.

Cosine similarity against all 858 ICDA-8 codes. The highest score for these 52 runs from 0.883 to 0.985, against 0.865 to 1.000 for the 302 codes that do have a match. The three highest and the three lowest are below as examples, and I have the other 46 if they are useful.

Code	Label	Average	Min	25th	Median	75th	Max	Closest target code
164	malignant neoplasm of thymus heart and mediastinum	0.813	0.584	0.776	0.810	0.846	0.985	142, malignant neoplasm of salivary gland
175	malignant neoplasm of male breast	0.816	0.578	0.782	0.816	0.848	0.984	174, malignant neoplasm of breast
237	neoplasm of uncertain behavior of endocrine glands and nervous system	0.821	0.594	0.785	0.822	0.855	0.984	238, neoplasm of unspecified nature of eye brain and other parts of nervous system
312	disturbance of conduct not elsewhere classified	0.775	0.678	0.751	0.775	0.797	0.895	306, special symptoms not elsewhere classified
625	pain and other symptoms associated with female genital organs	0.762	0.540	0.727	0.768	0.807	0.893	781, other symptoms referable to nervous system and special senses
315	specific delays in development	0.777	0.671	0.752	0.776	0.800	0.883	066, late effects of viral encephalitis

On both crosswalks the codes with no match sit inside the range of the codes that do, so no cut-off on either signal separates them without also losing a large share of the correct mappings.